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Clutch and work machine having same

Abstract

The present invention provides a clutch and a work machine having same. The clutch comprises an input member for receiving a driving force and rotating, an output member for outputting rotational motion, and at least one transmission member. The input member has at least one radially extending protrusion. The output member has at least one transmission engagement part. The transmission member is movable between a first position of engagement with the transmission engagement part and a second position of disengagement, being located in a path of rotation of the protrusion, and able to abut the protrusion and be driven to move to the first position. Displacement of the transmission member between the first position and second position has a radial component. According to the technical solution of the present invention, the function of automatically switching between transmission states according to the operating state of a prime mover can be achieved. The fact that displacement of the transmission member comprises a radial component helps to reduce the axial thickness of the clutch, so that the structure thereof is more compact.

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Background/Summary

(1) This application claims the benefit of priority to Chinese Patent Application No. CN 202210028937.4, filed on Jan. 11, 2022, the disclosure of which is incorporated by reference herein in its entirety.
TECHNICAL FIELD
(2) The present invention relates to a clutch and a work machine having same.
BACKGROUND ART
(3) Walk-behind work machines such as lawnmowers are generally provided with wheels for movement. The wheels enable the operator to move the walk-behind work machine to different work sites to perform corresponding work tasks. In some application scenarios, the operator may provide the driving force needed to move the work machine. In other application scenarios, the driving force may also be provided by a primer mover (such as a motor or internal combustion engine) provided on the work machine, the prime mover being coupled to the wheels by means of a transmission apparatus (such as a gearbox comprising a clutch), to realize transmission of the driving force, in which case the work machine may have a self-propulsion function.
(4) When the driving force is provided by a prime mover, it is desirable that the transmission apparatus be able to provide a stable and efficient transmission coupling. When an external traction force provided by the operator or another external apparatus serves as the driving force (in particular in a state in which the prime mover has stopped operating), it is desirable that the transmission apparatus be able to break the transmission coupling between the prime mover and the wheels, to prevent the prime mover from rotating with the wheels and causing a large amount of resistance. At present, walk-behind work machines are generally provided with a control apparatus such as a control rod and pull wire; the operator can switch manually between states of the transmission apparatus by means of the control apparatus, to establish or break the transmission coupling according to different operating modes. Such a manner of operation requires manual intervention and is quite complex, and is unable to achieve automatic switching between states of the transmission apparatus according to the operating state of the prime mover.
(5) Thus, there is a need to provide an alternative solution to at least partially alleviate or mitigate the abovementioned shortcomings.
SUMMARY OF THE INVENTION
(6) The objective of the present invention is to provide a clutch and a work machine having same, to achieve automatic switching between transmission states of a transmission apparatus according to the operating state of a prime mover while providing other additional advantages, such as the structure of the transmission apparatus being smaller and more compact, realizing a differential

function, etc.
(7) A clutch according to one aspect of the present invention comprises: an input member, the input member being configured to receive a driving force and rotatable about a rotation axis under the action of the driving force, the input member having at least one protrusion extending in a radial direction relative to the rotation axis; an output member, the output member being configured to output rotational motion and having at least one transmission engagement part; at least one transmission member, the at least one transmission member being configured to be movable between a first position in which it is engageable with the transmission engagement part and a second position of disengagement, the at least one transmission member being located in a path of rotational motion of the protrusion, and able to abut the protrusion and be driven to move to the first position, wherein displacement of the at least one transmission member between the first position and the second position has a component in the radial direction.
(8) In some embodiments, the clutch further comprises a biasing member, which biases the at least one transmission member towards the second position.
(9) In some embodiments, the biasing member is a spring, the spring having one end coupled to the at least one transmission member, and another end coupled to the input member or a mounting member for mounting the at least one transmission member.
(10) In some embodiments, the at least one transmission member is able to abut the protrusion and be driven to move to the second position.
(11) In some embodiments, a sharp bump is provided at an inner side of the at least one transmission member, the sharp bump being able to abut the protrusion when the input member rotates in a direction opposite to a drive direction.
(12) In some embodiments, the clutch is configured such that the input member rotates through a predetermined angle in a direction opposite to a drive direction when the outputting of rotational motion is stopped.
(13) In some embodiments, the at least one transmission member moves in a plane perpendicular to the rotation axis.
(14) In some embodiments, the at least one transmission member moves in the plane between the first position and the second position along a straight line inclined relative to the radial direction.
(15) In some embodiments, the protrusion comprises three protrusions arranged at uniform intervals in the circumferential direction about the rotation axis, and the at least one transmission member comprises three transmission members arranged in one-to-one correspondence with the protrusions.
(16) In some embodiments, the transmission engagement part of the output member comprises a stop wall extending parallel to the rotation axis, the at least one transmission member abutting the stop wall to achieve engagement with the transmission engagement part.
(17) In some embodiments, the stop wall extends substantially in the radial direction.
(18) In some embodiments, when the at least one transmission member is located at the second position, the at least one transmission member is located outside a path of rotational motion of the stop wall.
(19) In some embodiments, the output member is provided with a cam surface extending parallel to the rotation axis; the cam surface is located between two adjacent stop walls, and connects a radially outside end of one of the stop walls to a radially inside end of the other adjacent stop wall, at least part of the cam surface being a curved surface.
(20) In some embodiments, in the direction of the driving force, the angular velocity of rotation of the output member is greater than or equal to the angular velocity of rotation of the input member.
(21) In some embodiments, the clutch further comprises a mounting member for mounting the at least one transmission member, the mounting member being provided with an elongated slot defining a movement path of the at least one transmission member between the first position and the second position.

(22) In some embodiments, the elongated slot comprises a curved surface, to allow the at least one transmission member to pivot in the elongated slot.
(23) In some embodiments, the output member is provided with a recess, and at least one of the at least one transmission member, the mounting member and the input member is at least partially disposed in the recess; or the input member is provided with a recess, and at least one of the at least one transmission member, the mounting member and the output member is at least partially disposed in the recess.
(24) In some embodiments, the mounting member is provided with a mounting slot formed as a sunken cavity, the input member being accommodated in the sunken cavity.
(25) In some embodiments, the mounting member is provided with a damping element, the damping element being configured to be able to provide damping so that the mounting member tends to remain stationary.
(26) In some embodiments, the damping element provides damping by rubbing against a member other than the input member and the transmission member, in particular by rubbing against a transmission case accommodating the clutch.
(27) In some embodiments, the damping element is made of a rubber or silicone rubber material, and extends radially outwards relative to the mounting member.
(28) A work machine according to another aspect of the present invention comprises a prime mover, at least one wheel and the clutch as described above, the prime mover driving the input member of the clutch, and the output member of the clutch driving at least one of the at least one wheel to rotate.
(29) In some embodiments, the work machine comprises two clutches arranged in a mirror-image fashion, each clutch being coupled to at least one corresponding wheel.
(30) In some embodiments, the work machine is a gardening tool, in particular a lawnmower or a snow blower.
(31) The clutch and work machine according to the present invention are able to achieve the following beneficial effects: 1) The transmission member can move automatically to the first position of engagement with the output member under the action of the radially extending protrusion of the input member, and can move automatically to the second position of disengagement from the output member under the action of the biasing member when the input member stops moving. The function of automatically switching between transmission states according to the operating state of the prime mover is thereby achieved. 2) Displacement of the transmission member between the first position and second position comprises a radial component, helping to reduce the axial thickness of the clutch along the rotation axis, so that the structure of the clutch is more compact. 3) In some preferred embodiments, a speed differential function can be achieved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) To better understand the above and other objectives, features, advantages and functions of the present invention, the preferred embodiments shown in the drawings may be referred to. In the drawings, identical reference labels denote identical components. Those skilled in the art should understand that the drawings are intended to illustrate preferred embodiments of the present invention schematically, and have no limiting effect on the scope of the present invention, and the various components in the drawings are not drawn to scale.
(2) FIG. **1** is a 3D drawing of a work machine according to a preferred embodiment of the present invention.
(3) FIG. **2** is a 3D drawing of the work machine shown in FIG. **1**, viewed from another angle.

(4) FIG. **3** is a 3D drawing of a clutch according to a preferred embodiment of the present invention.
(5) FIG. **4** is a 3D drawing of the clutch shown in FIG. **3**, viewed from another angle.
(6) FIG. **5** is a schematic drawing of the clutch shown in FIG. **3** in a disengaged state, with the mounting member omitted.
(7) FIG. **6** is a schematic drawing of the clutch shown in FIG. **3** in an engaged state, with the mounting member omitted.
(8) FIG. **7** is a rear view of the clutch shown in FIG. **3**, with the output member omitted, showing a biasing member configured as a spring.
(9) FIG. **8** is a front view of a clutch according to another preferred embodiment of the present invention.
(10) FIG. **9** is a 3D view of a clutch according to another preferred embodiment of the present invention.
(11) FIG. **10** shows a transmission path of a work machine according to the present invention, from the prime mover to the clutch; and
(12) FIG. **11** is part of a sectional view taken along line A-A in FIG. **10**.

KEY TO THE DRAWINGS

(13) **1** lawnmower **10** base **11** cutting tool **12** wheel **13** hand support frame **20** prime mover **201** output shaft **21** transmission apparatus **211** first gear **212** second gear **213** third gear **214** fourth gear **215** fifth gear **216** coupling shaft **22** transmission case **30/30′/30″** clutch **31/31′** input member **311** input engagement part **312** protrusion **32/32′** output member **321** output engagement part **322** transmission engagement part **323** cam surface **323***a* downstream end **323***b* upstream end **324** recess **33/33′** transmission member **331** pin structure **332** sharp bump **34/34″** mounting member **341** elongated slot **342** fastener **343** first mounting slot **344** second mounting slot **345** damping element **346** holding part **35** spring **40** drive shaft AX rotation axis F drive direction F**1** direction opposite to drive direction

DETAILED DESCRIPTION OF EMBODIMENTS

(14) Particular embodiments of the present invention are now described in detail with reference to the drawings. The embodiments described here are merely preferred embodiments of the present invention. Based on these preferred embodiments, those skilled in the art will be able to think of other ways in which the present invention could be implemented, all of which likewise fall within the scope of the present invention.
(15) The present invention provides a work machine, in particular a gardening tool, which is able to move between different positions by means of wheels under the action of an external traction force and/or a driving force provided by its own prime mover. In addition, the present invention further provides a clutch for the work machine. The clutch is disposed between the prime mover and the wheels, and used to transmit the driving force of the prime mover to the wheels.
(16) The work machine is preferably provided with a hand support frame, to facilitate gripping and application of force by the operator. Such a work machine may also be called a walk-behind work machine. One example of a walk-behind work machine is a lawnmower, which is configured to be able to move over the growth surface of grass or a lawn and be operated to perform a grass-cutting function. Such an action is usually referred to as “trimming a lawn”, and is generally performed by a gardener or landscape worker to maintain the surface of the lawn. In addition, other examples of walk-behind work machines may be snow blowers, plowing machines, micro tillers and wheeled vacuum cleaners. Preferred embodiments according to the present invention are presented in detail below with reference to the drawings.
(17) FIGS. **1** and **2** show a lawnmower **1** as an example of the work machine according to the present invention, at least comprising a base **10**, a cutting tool **11**, wheels **12**, a hand support frame **13**, a prime mover **20** and a transmission apparatus **21**. The cutting tool **11** is disposed at the bottom of the base **10**, and can perform a job under the action of a driving force provided by the prime

mover **20**, to cut and clear weeds on a lawn. The wheels **12** are used to support the lawnmower **1** on a working surface (such as a lawn), and can rotate under the action of a driving force provided by the prime mover **20**, to move the lawnmower **1** between different positions. The same prime mover **20** may be used to provide the driving forces to the wheels **12** and the cutting apparatus **11** respectively, or different prime movers **20** may be provided to provide the driving forces separately. In the embodiment shown in the figures, the lawnmower **1** comprises four wheels **12**, arranged opposite each other at left and right sides of a front end and a rear end of the base **10** respectively, to realize stable support. Of course, in other embodiments, the number of wheels **12** may be set to one, two, three or more as required. The hand support frame **13** is disposed at a rear part of the lawnmower **1** and is angled upwards obliquely towards the rear, so that it can be conveniently gripped by the operator in the course of working to push the lawnmower **1** forwards and backwards.

(18) Referring to FIG. **2**, the prime mover **20** is disposed at the rear end of the base **10**, close to the wheels **12** located at the rear end. The transmission apparatus **21** is located between the prime mover **20** and the wheels **12**, and used to transmit a driving force therebetween. An example of the prime mover **20** could be an electric motor or internal combustion engine, etc., which may be configured to provide a driving force to only one wheel **12**, or may provide a driving force simultaneously to two or more wheels **12**. It is also possible to provide more than one primer mover **20** and transmission apparatus **21**, to drive multiple wheels **12** separately. In the case where the prime mover **20** provides a driving force, a driven wheel **12** may be called a driving wheel, and a wheel **12** which rotates passively as the lawnmower **1** moves may be called a driven wheel. In the embodiment shown, the two wheels **12** close to the rear end of the base **10** are driven by the prime mover **20**, so are driving wheels. The two wheels close to the front end of the base **10** are driven wheels.

(19) FIG. **10** shows the transmission path from the prime mover **20** to the driving wheels via the transmission apparatus **21**. In some embodiments, the transmission apparatus **21** comprises a transmission gear train and a clutch. An output shaft of the prime mover **20** is transmission-coupled to the clutch **30** via the transmission gear train, and further transmission-coupled to the driving wheels via the clutch **30**. In the embodiment shown, an extremity of the output shaft **201** of the prime mover **20** is configured in the form of a gear, and meshed with a first gear **211** of the transmission gear train. The output shaft **201** and the first gear **211** have substantially the same linear velocity. However, the diameter of the first gear **211** is greater than the diameter of the output shaft **201**, so the speed outputted by the prime mover **20** is reduced a first time at the first gear **211**. The first gear **211** and a second gear **212** are arranged coaxially, and have substantially the same angular velocity. The second gear **212** and a third gear **213** are meshed, and have substantially the same linear velocity, but the diameter of the third gear **213** is greater than the diameter of the second gear **212**. Thus, the speed outputted by the prime mover **20** is reduced a second time at the third gear **213**. The third gear **213** is further arranged coaxially with a fourth gear **214**. The fourth gear **214** and a fifth gear **215** are meshed, and have substantially the same linear velocity, but the diameter of the fifth gear **215** is greater than the diameter of the fourth gear **214**. Thus, the speed outputted by the prime mover **20** is reduced a third time at the fifth gear **215**. It will be understood that in different embodiments, it is possible to flexibly choose the number of stages in the transmission gear train and whether the transmission result is a reduction in speed or an increase in speed, according to actual needs.

(20) As shown in FIG. **11**, the fifth gear **215** forms a final-stage gear of the transmission gear train. Furthermore, the transmission gear train further comprises a coupling shaft **216** arranged coaxially with the fifth gear **215**. Optionally, the final-stage gear of the transmission gear train is transmission-coupled to two clutches **30** simultaneously. These two clutches **30** are arranged substantially in a mirror-image fashion relative to the final-stage gear. Output ends of the two clutches **30** are coupled to respectively corresponding drive shafts **40**. An extremity of each drive

shaft **40** is further coupled to one driving wheel (not shown) respectively. For example, the extremity of the drive shaft **40** may be coupled to the driving wheel directly via a pin or splines, etc. Alternatively, a gear structure may be provided at the extremity of the drive shaft **40**, and a ring gear may be provided on the driving wheel, with the transmission coupling being accomplished by meshing of the gear structure with the ring gear. In this way, the lawnmower **1** is able to drive the two driving wheels at the left and right sides simultaneously by means of a single prime mover **20** cooperating with a single transmission apparatus **21**. The transmission gear train and the clutches **30** form the main part of the transmission apparatus **21**. Furthermore, a transmission case **22** is also provided, in which the transmission gear train and the clutches **30** are accommodated, and which serves a protecting function, preventing the ingress of debris while also providing lubrication for the transmission gear train and other structures. Furthermore, having the clutches arranged in a mirror-image fashion can allow a speed differential (described in detail below) to arise between the driving wheels respectively corresponding to the two clutches, thereby allowing the work machine to accomplish a turning action without the prime mover **20** stopping operation, thus making it easier to operate the work machine.

(21) FIGS. **3-6** show the clutch **30** according to some embodiments of the present invention, mainly comprising an input member **31**, an output member **32** and transmission members **33**.

(22) The input member **31** for example has an input engagement part **311** configured as an internal spline structure, and may be coupled to the prime mover **20** directly (not via a drive train) or indirectly (via a drive train) by means of the input engagement part **311**, thereby being able to receive a driving force from the prime mover **20**, and rotate around a rotation axis AX in a drive direction F under the action of the driving force. The output member **32** is used to output rotational motion to the outside, and for example has an output engagement part **321** configured as an internal spline structure, and may be coupled to at least one wheel **12** directly or indirectly by means of the output engagement part **321**, thereby being able to drive at least one wheel **12** to rotate. As shown in FIG. **10**, in some embodiments, the input engagement part **311** is coupled to the prime mover **20** via the transmission gear train, while the output engagement part **321** is coupled to the wheel **12** (driving wheel) via the drive shaft **40**.

(23) The transmission members **33** are movably arranged between the input member **31** and the output member **32**. When the transmission members **33** are located at a first position as shown in FIG. **6**, they can engage with the output member **32**, transmitting the rotational motion of the input member **31** to the output member **32**, such that the output member **32** rotates together with the input member **31**. At this time, the clutch **30** is in an engaged state. When the transmission members **33** move to a second position as shown in FIG. **5**, they disengage from the output member **32**, at which time there is no driving force acting on the output member **32** via the input member **31**, so the output member **32** remains stationary. The clutch **30** is in a disengaged state. The transmission members **33** may also be called movable pawls.

(24) It will be understood that during normal driving, the input member **31** drives the output member **32** to rotate by means of the transmission members **33**, in turn driving the wheel **12** to rotate by means of the drive shaft **40**. At this time, the input member **31** and the output member **32** have substantially the same angular velocity of rotation. As stated above, in some embodiments, two clutches **30** are transmission-coupled to the same final-stage gear, and arranged in a mirror-image fashion relative to the final-stage gear. Furthermore, the clutch **30** is configured to allow the angular velocity of rotation of the output member **32** to be greater than the angular velocity of rotation of the input member **31**. In this way, when the operator applies an external traction force to the wheel **12** at one side or applies unbalanced external traction forces to the wheels **12** at both sides, the wheel **12** at one side acted on by the larger force has a greater rotation speed than the wheel **12** at the other side acted on by the smaller force, i.e. a speed differential forms, at which time the work machine is able to accomplish a turning action. Since the angular velocity of rotation of the output member **32** of the clutch can be greater than the angular velocity of rotation of the

input member **31**, such an operation can automatically achieve a speed differential effect, without the need to manually switch the clutch **30** to the disengaged state. When the external traction force disappears, the clutch **30** will automatically re-enter the engaged state under the action of the driving force of the prime mover **20**.

(25) According to the present invention, when the driving force acts on the input member **31**, the clutch **30** can automatically switch to the engaged state without the need for active operation by the operator, to output rotational motion to the outside. Specifically, referring to FIGS. **5** and **6**, the input member **31** has protrusions **312** extending outwards in radial directions relative to the rotation axis AX. The transmission members **33** are disposed in the path of rotational motion of the protrusions **312**. When the driving force acts on the input member **31**, the protrusions **312** rotate around the rotation axis AX, and thereby contact the transmission members **33**. As the rotational motion continues, the transmission members **33** are driven to move to the first position shown in FIG. **6** under the action of the protrusions **312**.

(26) The output member **32** is provided with transmission engagement parts **322**. In the embodiment shown, the transmission engagement parts **322** are configured in the form of stop walls extending substantially parallel to the rotation axis AX. When the transmission members **33** move to the first position, they abut the stop walls, thereby engaging with the transmission engagement parts **322**. Thus, the transmission members **33** transmit the action force of the protrusions **312** to the transmission engagement parts **322**, so that the output member **32** rotates together with the input member **31**. The clutch **30** automatically switches to the engaged state. Preferably, the stop walls extend substantially in said radial directions.

(27) In some embodiments, the clutch **30** is further provided with biasing members, which apply a biasing force to the transmission members **33** to bias them towards the second position. As shown in FIG. **7**, the biasing member may for example a spring **35**, having one end coupled to the transmission member **33**, and another end coupled to the protrusion **312** which drives the transmission member **33** to move towards the first position. Preferably, the clutch **30** is further configured such that when the outputting of rotational motion is stopped, the input member **31** rotates through a predetermined angle in a direction F**1** (see FIG. **8**) opposite to the drive direction. This configuration may be accomplished by, for example, causing the prime mover to rotate in reverse for a predetermined time (e.g. 0.1 s) or through a predetermined angle after ceasing to operate. Thus, after the prime mover has ceased to operate, the input member **31** rotates in reverse through a predetermined angle, a sufficient gap arises between the protrusion **312** and the transmission member **33**, and the transmission member **33** can move from the first position to the second position shown in FIG. **5** under the action of the biasing member, thereby causing the clutch **30** to automatically switch to the disengaged state.

(28) In some other embodiments, the biasing member may be omitted, and the transmission member may be moved to the second position by the protrusion of the input member abutting the transmission member. Referring to the clutch **30′** shown in FIG. **8**, a sharp bump **332** is provided on an inner side of the transmission member **33′**. When the outputting of rotational motion is stopped, the input member **31′** rotates through a predetermined angle in the direction F**1** opposite to the drive direction. In the course of rotating in reverse, the protrusion **312′** abuts the sharp bump **332** of the adjacent transmission member **33′** located behind (i.e. upstream of) the protrusion in the drive direction F. As the rotation in reverse continues, the protrusion **321′** drives the transmission member **33′** to move to the second position. It will be understood that with such a configuration, the force applied to the sharp bump **332** by the protrusion **312′** is in a direction tangential to the direction of rotation thereof, so might cause the transmission member **33′** to pivot in the process of moving towards the second position.

(29) In the second position, the transmission members **33** are located completely outside the path of rotational motion of the transmission engagement parts **322**. At this time, the output member **32** can freely rotate independently of the input member **31** and the transmission members **33**. In other

words, if the operator wishes that the lawnmower **1** be driven to move by an external traction force (such as a pushing or pulling force applied by the operator or another external driving means) when the prime mover **20** has stopped operating, the wheel **12** coupled to the output member **32** of the clutch **30** need only drive the output member **32** to rotate with it. No action force will arise between the output member **32** and the transmission members **33**/input member **31** or transmission gear train/prime mover **20**. It is thus possible to reduce resistance when an external traction force drives the lawnmower **1**. Moreover, at this time the wheel **12** can also drive the output member **32** to rotate in reverse, without interfering with the transmission members **33** or the input member **31**. That is to say, the operator can pull the lawnmower **1** backwards.

(30) As shown in FIG. **4**, preferably, the clutch **30** further comprises a mounting member **34**, configured as a disc-shaped structure and provided with elongated slots **341**. Correspondingly, the transmission members **33** are provided with protruding pin structures **331**. The transmission members **33** may be mounted on the mounting member **34** by insertion of the pin structures **331** into the elongated slots **341**. Preferably, the elongated slot **341** is a penetrating slot, and a fastener **342** comprising a spacer and a screw is provided at an end of the pin structure **331**. The size of at least the spacer is greater than the width of the elongated slot **341**, to prevent the pin structure **331** from coming out of the elongated slot **341**. It will be understood that in another embodiment, it is also possible to have the elongated slot provided in the transmission member, and have the pin structure cooperating therewith provided on the mounting member.

(31) In addition, in an embodiment in which a biasing member is used to drive the transmission member, the mounting member may also be used to provide a force application point for the biasing member configured as a spring. One end of the spring is still coupled to the transmission member, while the other end is coupled to the mounting member, not to the protrusion of the input member as described above. The specific form of the spring may be chosen flexibly as required, e.g. a tension spring, a torsion spring, a coil spring, a specially-shaped spring, etc.

(32) When the transmission member **33** moves between the first position and the second position, the pin structure **331** moves in the length direction of the elongated slot **341**. Thus, the elongated slot **341** defines the path of movement of the transmission member **33** between the first position and the second position. According to the present invention, the movement path of the transmission member **33** has a component in a radial direction relative to the rotation axis AX. That is to say, movement of the transmission member **33** between the first position and the second position causes a change in the radial position thereof relative to the rotation axis AX. Preferably, the movement path of the transmission member **33** lies completely within a plane perpendicular to the rotation axis AX. That is to say, movement of the transmission member **33** between the first position and the second position will not give rise to axial displacement along the rotation axis AX. Such a configuration helps to reduce the axial thickness of the clutch **30** along the rotation axis AX, and will not give rise to a change in the axial dimension during operation, so makes the structure thereof compact.

(33) Further preferably, the path of movement of the transmission member **33** between the first position and the second position is a straight line which is inclined relative to a radial direction. For example, the movement path may be substantially in a direction tangential to the path of rotational motion of the protrusion **312**, or may deviate from this tangential direction by a small angle; in this way, the action torque applied to the transmission member **33** by the protrusion **312** can be maximized. Moreover, correspondingly, it is only necessary for the elongated slot **341** to be configured as a straight slot, so the structure is simple. However, it will be understood that the path of movement of the transmission member **33** between the first position and the second position may also be configured as a curve as required, i.e. the elongated slot **341** is a curved slot. In addition, the elongated slot may comprise a curved surface, to allow pivoting of the transmission member in the process of moving towards the second position, in an embodiment in which the protrusion is used to drive the transmission member towards the second position.

(34) Referring to FIG. **7**, the mounting member **34** is provided with first mounting slots **343**. The first mounting slots **343** are substantially configured to be elongated in shape, and are open at a circumferential side of the mounting member **34**. The transmission members **33** are accommodated in the first mounting slots **343**, and are capable of moving along the first mounting slots **343** to the first position in which they project through the openings thereof, and retreating to the second position. Thus, the first mounting slots **343** can also serve to define the movement paths of the transmission members **33**. In addition, a substantially round second mounting slot **344** is further provided in a middle region of the mounting member **34**, the second mounting slot communicating with the first mounting slots **343**. The input member **31** is accommodated in the second mounting slot **344** and is freely rotatable in the second mounting slot **344**. Such a configuration can further reduce the axial thickness of the clutch **30**.
(35) It will be understood that if there is an action force between the transmission member and the mounting member, for example a frictional force, then when the input member pushes the transmission member, the latter might be unable to overcome the action force to move relative to the mounting member to the first position, and will instead directly drive the mounting member to rotate under the action of the action force. Since the transmission member has not moved to the first position, it will not engage with the output member, and the clutch will idle in the disengaged state.
(36) To avoid this problem, referring to FIG. **9**, in the clutch **30**″ according to some embodiments, the mounting member **34**″ is provided with damping elements **345**, which can provide damping for the mounting member **34**″, so that it tends to remain stationary. The damping elements **345** may be entirely or partially made of a rubber or silicone rubber material with a large coefficient of friction, may extend radially outwards relative to the mounting member **34**″, and are in contact with an inner wall of the transmission case mentioned above for example. In addition, the mounting member **34**″ is provided with holding parts **346** for mounting the damping elements **345**. When the input member rotates under the driving action of the prime mover, the protrusions of the input member abut the transmission members and push them to move towards the first position. At this time, the damping elements **345** rub against the inner wall of the transmission case, providing damping for the mounting member **34**″, so that it tends to remain stationary. In this way, the transmission member is able to overcome the action force between the transmission member and the mounting member **34**″ under the pushing action of the input member, and move smoothly relative to the mounting member **34**″ to the first position, so as to engage with the output member.
(37) As shown in FIG. **5**, the output member **32** is provided with a recess **324** sunk along the rotation axis AX; at least part of at least one of the transmission members **33**, the mounting member **34** and the input member **31** is disposed in the recess **324**. Such a configuration can likewise further reduce the axial thickness of the clutch **30** along the rotation axis AX. In another embodiment, when the input member has a larger radial dimension and the output member has a smaller radial dimension, it is also possible to have the recess provided in the input member, and have the transmission members, the mounting member and the output member at least partially disposed in the recess.
(38) It will be understood that at least one protrusion **312**, at least one transmission member **33** and at least one transmission engagement part **322** may be provided, to enable transmission of the driving force. In the embodiment shown, the clutch **30** is provided with three sets of protrusions **312**, transmission members **33** and transmission engagement parts **322** which are in one-to-one correspondence and cooperate with each other; this three-set structure is distributed at uniform intervals in the circumferential direction about the rotation axis AX. However, the numbers of cooperating protrusions, transmission members and transmission engagement parts may be chosen flexibly according to changes in the dimensions and structure of the clutch; moreover, the protrusions and transmission members are not necessarily in a one-to-one correspondence relationship with the transmission engagement parts, as long as transmission of the driving force

can be achieved. For example, in some embodiments, the number of transmission engagement parts may be twice that of the protrusions and transmission members, etc.
(39) In some embodiments, the output member **32** is further provided with cam surfaces **323**, configured as faces that face the transmission members **33** and are substantially parallel to the rotation axis AX, and each having a downstream end **323***a* and an upstream end **323***b* in the drive direction F, these two ends being respectively connected to two adjacent transmission engagement parts **322**. Specifically, in the embodiment shown in which the transmission members **33** are located at a radially inner side of the transmission engagement parts **322**, the downstream end **323***a* of the cam surface **323** is connected to a radially outside end of the corresponding stop wall, while the upstream end **323***b* of the cam surface **323** is connected to a radially inside end of the corresponding stop wall (i.e. the adjacent stop wall located behind the stop wall connected to the downstream end **323***a* in the drive direction F). Preferably, at least part of the cam surface **323** is a smooth curved surface.
(40) As an alternative embodiment, when the transmission members are located at a radially outer side of the transmission engagement parts, the downstream end of the cam surface should be connected to a radially inside end of the corresponding stop wall, while the upstream end of the cam surface should be connected to a radially outside end of the corresponding stop wall.
(41) When the prime mover **20** is in an operational state, if an external traction force is applied at the same time, for example the lawnmower **1** is pushed manually, then the rotation speed of the wheels **12** increases, and the output members **32** coupled to the drive shafts **40** are driven via the drive shafts to rotate at a faster speed. The stop wall structure described above allows the output member **32** to rotate at a higher angular velocity than the input member **31** and transmission members **33**. That is to say, the angular velocity of the stop walls is greater than the angular velocity of the transmission members **33** in abutment therewith, so the transmission members **33** move backwards relative to the abutted stop walls and thereby disengage, and slide along the cam surfaces **323**. The cam surfaces **323** may press the transmission members **33** back to the second position, such that the clutch **30** automatically switches to the disengaged state. Thus, even if the transmission members **33** are located in the path of rotational motion of the transmission engagement parts **322** at this time, they will not cause interference to relative motion between the output member **32** and the transmission members **33**.
(42) Such a configuration allows the output member **32** to move at a higher speed relative to the input member **31** and the transmission members **33**. For example, in the process of the lawnmower **1** being driven by the prime mover **20**, when it is necessary to turn, the operator can apply a pushing force, so that the wheel **12** of the lawnmower **1** that is located at the outer side of the turning radius rotates at a faster speed, and a speed differential thus arises between the wheels **12** at the inner and outer sides of the turning radius, to smoothly accomplish the turning action. Due to the presence of the clutch **30**, even if the operator does not stop the prime mover **20**, it will not cause any interference or effect on the driving of the wheel **12** located at the outer side of the turning radius—this is very convenient. When turning is complete, the operator stops applying the pushing force, the rotation speed of the wheel **12** falls until it is equal to or slightly less than the rotation speed of the input member **31** and transmission members **33** of the clutch **30**, the transmission members **33** are pushed to the first position by the input member **31**, the clutch **30** automatically switches to the engaged state, and the lawnmower **1** can thus return to the pre-turning state of being driven by the prime mover **20**. The clutch **30** is able to automatically switch between states to adapt to changes in the rotation speed of the wheels **12**, without any need to manually switch the state of the clutch **30**.
(43) The above description of various embodiments of the present invention is provided to a person skilled in the art for descriptive purposes. It is not intended that the present invention be exclusively or limited to a single disclosed embodiment. As mentioned above, those skilled in the art will understand various alternatives and variations of the present invention. Thus, although

some alternative embodiments have been specifically described, those skilled in the art will understand, or develop with relative ease, other embodiments. The present invention is intended to include all alternatives, modifications and variants of the present invention described here, as well as other embodiments which fall within the spirit and scope of the present invention described above.

Claims

1. A clutch, comprising: an input member, the input member being configured to receive a driving force and rotatable about a rotation axis under action of the driving force, the input member having at least one protrusion extending in a radial direction relative to the rotation axis; an output member, the output member being configured to output rotational motion and having at least one transmission engagement part; and at least one transmission member, the at least one transmission member being configured to be movable between a first position in which it is engageable with the transmission engagement part and a second position of disengagement, the at least one transmission member being located in a path of rotational motion of the protrusion, and able to abut the protrusion and be driven to move to the first position, wherein displacement of the at least one transmission member between the first position and the second position has a component in the radial direction, and wherein the at least one transmission member configured to disengage the at least one protrusion in the second position.
2. The clutch according to claim 1, wherein the clutch further comprises a biasing member, which biases the at least one transmission member towards the second position.
3. The clutch according to claim 2, wherein the biasing member is a spring, the spring having one end coupled to the at least one transmission member, and another end coupled to the input member or a mounting member for mounting the at least one transmission member.
4. The clutch according to claim 2, wherein the clutch is configured such that the input member rotates through a predetermined angle in a direction opposite to a drive direction by the biasing spring when the outputting of rotational motion is stopped.
5. The clutch according to claim 1, wherein the at least one transmission member is able to abut the protrusion and be driven to move to the second position.
6. The clutch according to claim 5, wherein a sharp bump is provided at an inner side of the at least one transmission member, the sharp bump being able to abut the protrusion when the input member rotates in a direction opposite to a drive direction.
7. The clutch according to claim 1, wherein the at least one transmission member moves in a plane perpendicular to the rotation axis.
8. The clutch according to claim 7, wherein the at least one transmission member moves in the plane between the first position and the second position along a straight line inclined relative to the radial direction.
9. The clutch according to claim 1, wherein the at least one protrusion comprises three protrusions arranged at uniform intervals in a circumferential direction about the rotation axis, and the at least one transmission member comprises three transmission members arranged in one-to-one correspondence with the protrusions.
10. The clutch according to claim 1, wherein the transmission engagement part of the output member comprises a stop wall extending parallel to the rotation axis, the at least one transmission member abutting the stop wall to achieve engagement with the transmission engagement part.
11. The clutch according to claim 10, wherein the stop wall extends substantially in the radial direction.
12. The clutch according to claim 10, wherein when the at least one transmission member is located at the second position, the at least one transmission member is located outside a path of rotational motion of the stop wall.

13. The clutch according to claim 10, wherein the output member is provided with a cam surface extending parallel to the rotation axis; and wherein the cam surface is located between two adjacent stop walls, and connects a radially outside end of one of the stop walls to a radially inside end of the other adjacent stop wall, at least part of the cam surface being a curved surface.
14. The clutch according to claim 1, wherein in the direction of the driving force, an angular velocity of rotation of the output member is greater than or equal to an angular velocity of rotation of the input member.
15. The clutch according to claim 1, wherein the clutch further comprises a mounting member for mounting the at least one transmission member, the mounting member being provided with an elongated slot defining a movement path of the at least one transmission member between the first position and the second position.
16. The clutch according to claim 15, wherein the mounting member is provided with a damping element, the damping element being configured to be able to provide damping so that the mounting member tends to remain stationary.
17. The clutch according to claim 16, wherein the damping element provides damping by rubbing against a member other than the input member and the transmission member.
18. The clutch according to claim 16, wherein the damping element is made of a rubber or silicone rubber material, and extends radially outwards relative to the mounting member.
19. The clutch according to claim 15, wherein the mounting member is provided with a mounting slot formed as a sunken cavity, the input member being accommodated in the sunken cavity.
20. The clutch according to claim 15, wherein the elongated slot comprises a curved surface, to allow the at least one transmission member to pivot in the elongated slot.
21. The clutch according to claim 15, wherein the output member is provided with a recess, and at least one of the at least one transmission member, the mounting member and the input member is at least partially disposed in the recess; or the input member is provided with a recess, and at least one of the at least one transmission member, the mounting member and the output member is at least partially disposed in the recess.
22. A work machine, comprising a prime mover, at least one wheel and the clutch according to claim 1, the prime mover driving the input member of the clutch, and the output member of the clutch driving at least one of the at least one wheel to rotate.
23. The work machine according to claim 22, wherein the clutch comprises two clutches arranged in a mirror-image fashion, each clutch being coupled to at least one corresponding wheel.
24. The work machine according to claim 22, wherein the work machine is a gardening tool.